Case Study 17: Student Performance Trend Analysis

1. Problem Statement

Analyze sequential test scores for a student across academic assessments to calculate key statistical metrics (average, highest, lowest) and automatically classify performance trajectory trends (improving, declining, or fluctuating).

2. Objectives

- Evaluate longitudinal student marks over sequential assessment periods.
- Derive statistical metrics including mean performance score, maximum score, and minimum score.
- Apply conditional logic loops to categorize academic trajectory patterns automatically.

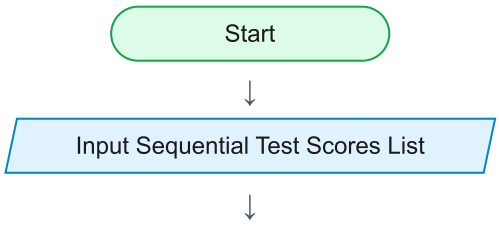
3. Assumptions

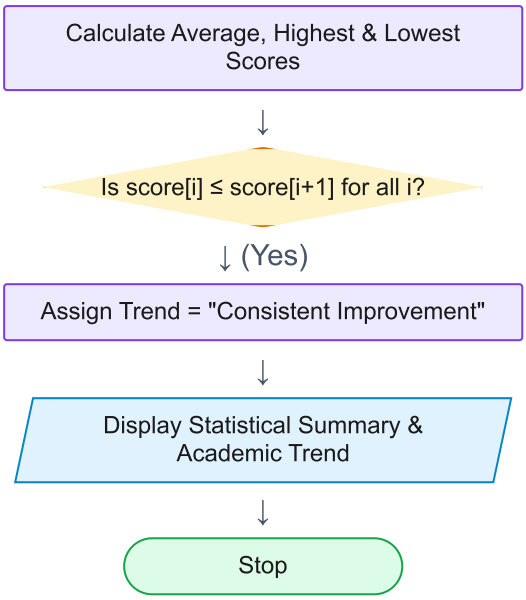
- Test scores are non-negative numeric values represented in chronological order.
- Assessments have a fixed score scale (e.g., 0 to 100 marks).
- Trend classification requires a minimum sequence of 2 or more assessment results.

4. Algorithm

1. Start.
2. Read list of sequential test scores.
3. Compute average score: `average = sum(scores) / length(scores)`.
4. Determine peak score `highest = max(scores)` and lowest score `lowest = min(scores)`.
5. Evaluate sequential trends:
   - If `scores[i] <= scores[i+1]` for all consecutive elements, set trend = **"Consistent Improvement"**.
   - Else if `scores[i] >= scores[i+1]` for all consecutive elements, set trend = **"Declining Trend"**.
   - Otherwise, set trend = **"Fluctuating Performance"**.
6. Display calculated summary statistics and identified performance trajectory.
7. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT scores IF length(scores) == 0 THEN PRINT "No scores provided." EXIT
ENDIF total = 0 highest = scores[0] lowest = scores[0] FOR i FROM 0 TO
length(scores) - 1 DO total = total + scores[i] IF scores[i] > highest THEN highest
= scores[i] ENDIF IF scores[i] < lowest THEN lowest = scores[i] ENDIF ENDFOR
avg_score = total / length(scores) is_improving = True is_declining = True FOR i
FROM 0 TO length(scores) - 2 DO IF scores[i] > scores[i+1] THEN is_improving =
False ENDIF IF scores[i] < scores[i+1] THEN is_declining = False ENDIF ENDFOR IF
is_improving THEN trend = "Consistent Improvement" ELSE IF is_declining THEN trend
= "Declining Trend" ELSE trend = "Fluctuating Performance" ENDIF PRINT "Average
Score:", avg_score PRINT "Highest Score:", highest PRINT "Lowest Score:", lowest
PRINT "Performance Trend:", trend STOP
```

7. Test Case and Expected Output

Test Case 1: Sequential Scores [65, 72, 78, 85, 90]

```
Student Performance Summary: - Average: 77.0 - Highest: 90 - Lowest: 65 - Trend:
Consistent Improvement
```

Test Case 2: Sequential Scores [88, 82, 75, 70]

```
Student Performance Summary: - Average: 78.75 - Highest: 88 - Lowest: 70 - Trend:
Declining Trend
```

8. Conclusion

The Student Performance Trend Analysis tool evaluates historical marks to provide comprehensive statistical metrics and automated trajectory categorization, assisting educators in identifying student progress patterns over time.